

Thermal Camera

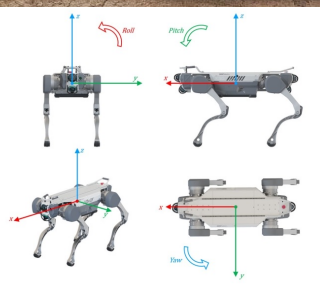
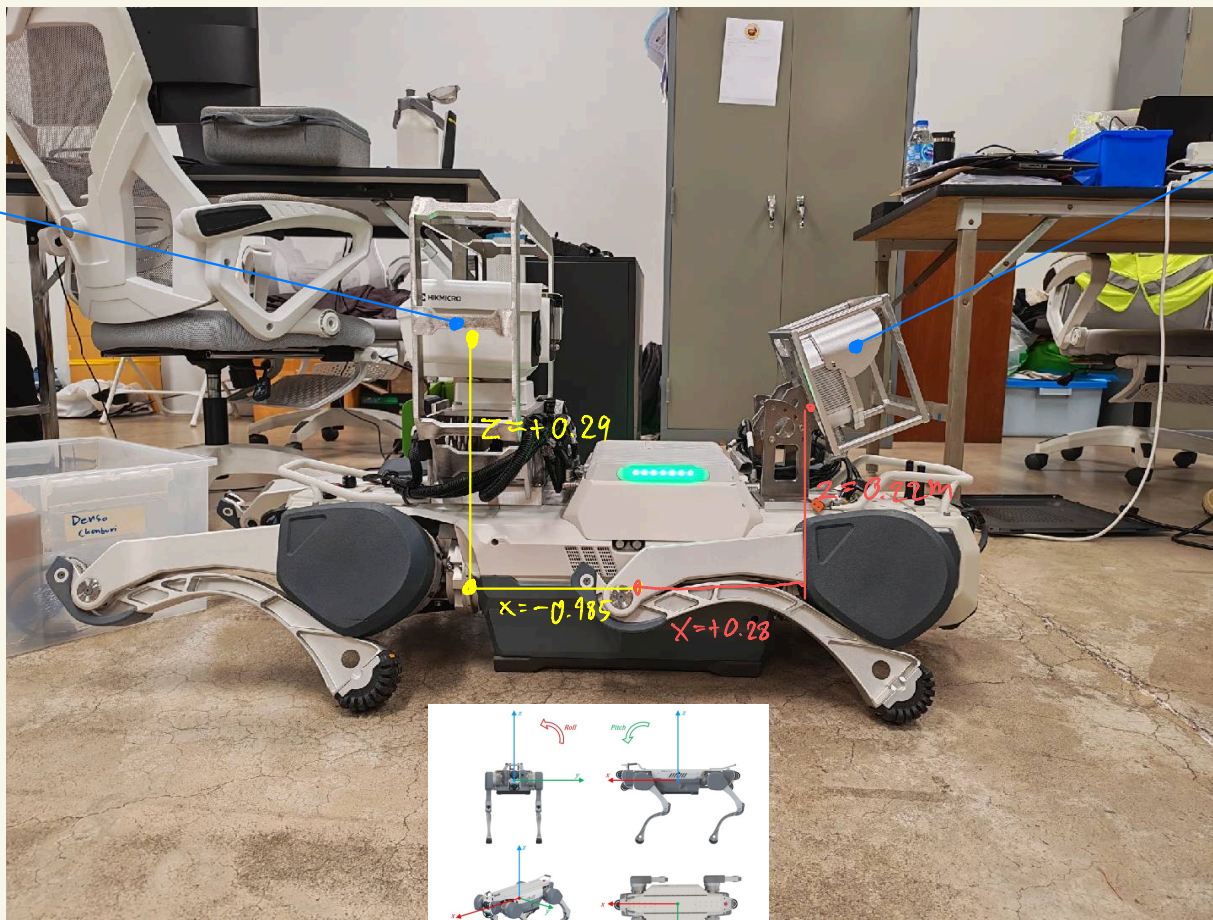
8.6 kg

$x = -0.18$ m
 $y = 0.0$ m
 $z = 0.28$ m

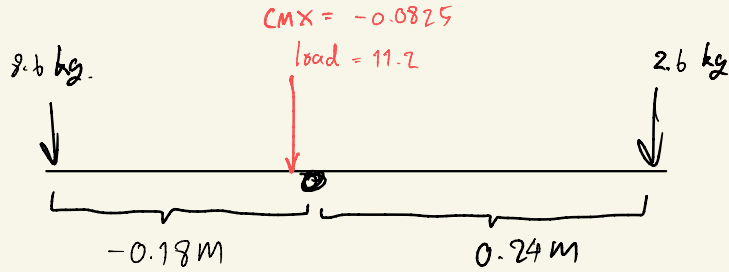
Acoustic Target

2.6 kg

$x = 0.24$ m
 $y = 0.0$ m
 $z = 0.23$ m



X axis



$$\begin{aligned}\text{Center of mass } X &= \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \\ &= \frac{8.6(-0.18) + 2.6(0.24)}{8.6 + 2.6} \\ &= -0.0825 \text{ m}\end{aligned}$$

$$y \text{ axis} = 0$$

Z axis

$$\begin{aligned}\text{Center of mass } z &= \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2} \\ &= \frac{8.6(0.28) + 2.6(0.23)}{8.6 + 2.6} \\ &= 0.2591 \text{ m}\end{aligned}$$